

Cloudflare for Hackathons

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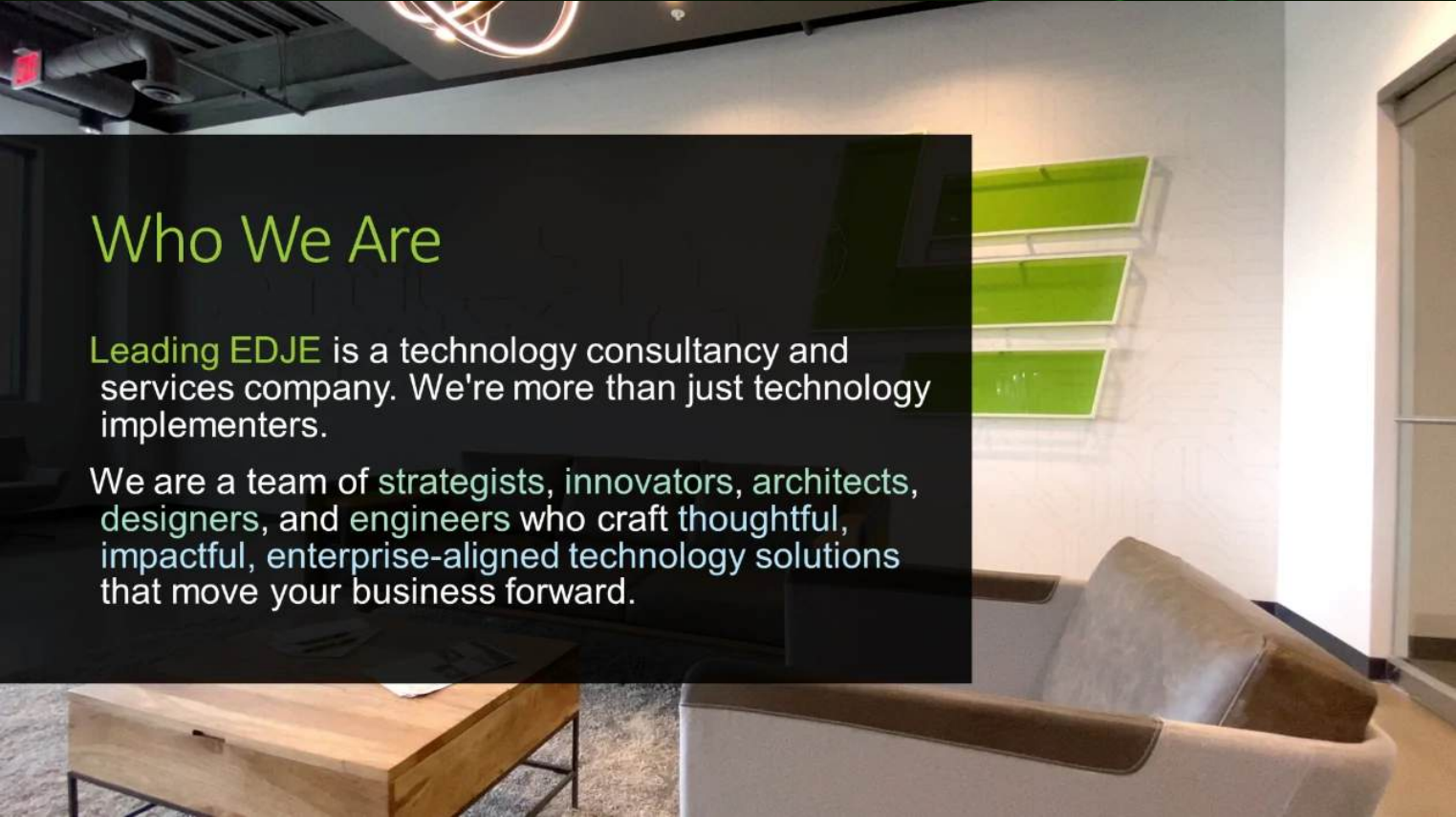
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Who We Are

Leading EDJE is a technology consultancy and services company. We're more than just technology implementers.

We are a team of strategists, innovators, architects, designers, and engineers who craft thoughtful, impactful, enterprise-aligned technology solutions that move your business forward.



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Hackathons: Introduction

Common hackathon constraints ...

- Very short timelines.
- Limited infrastructure resources.
- Need for quick iteration and fast demos.

Personal Implementations ...

- Tried **Angular** (took too long to initialize and pull down).
- Used **HTML**, **CSS**, & **jQuery-lite** (have experience with jQuery).
- Had others generate mock data (excel converted to **JSON**).

Thoughts ...

- Implementing features without framework can't scale past event.
- No "real" functionality (Create, Read, Update, & Delete).

We Are Here ...

Lwin & Bob

- Found common excitement.



Cloudflare

- Worker Scripts
- Key Vaults
- D1 Database

Super Script: Built a tool for managing ...

- Redirects
- Split-tests
- Dynamic Content

This script handles millions of Worker Script hits per day.

Overview

1. Accounts (with Users)
2. Domains
3. Pricing
4. Worker Scripts (APIs, CRON Jobs, Environment Variables, Secrets)
5. Key Vaults (fast document database)
6. D1 Database (SQLite)
7. Pages (repository or upload; frameworks)
8. ~~Image Storage (transformations)~~

Accounts & Domains

Accounts

- **Manage** > **Members** allows invitation of members.

Domains

- Manage
- Transfer
- Register



Pricing

Pricing (Workers plans: **Free**)

- **Worker & Pages Functions** (10ms CPU, 100k/day)
- **Builds** (1 concurrent slot)
- **D1** (100k writes/day, 5GB)
- **KV** (100k read/day, 1k write, delete, list/day)
- **Hyperdrive** (100k queries/day)

Pricing (Workers plans: **Paid**)

- **Worker & Pages Functions** (15min CPU, 10M/day)
- **Builds** (6 concurrent slots)
- **D1**
- **KV**
- **Hyperdrive** (unlimited)

Worker Scripts

Think "API." ...

- JavaScript
- TypeScript
- Python
- Rust

Findings ...

- 60-second cache.
- Connect: Routes, DB, KV, Secrets (and more).
- Free 100,000 requests/day (paid, no limit).
- Module Pattern (`import/export`, multiple files).
- `ctx.waitUntil` (extend lifetime of worker).
- Cron Jobs.

Keep in mind ...

- Cache-buster, CORS, paging.

Deployment

- GitHub Actions (build process).
- Connect to Github via Settings (was able to not build).

Options

- Sites deployed as Worker Scripts rather than Pages (this feature is the direction Cloudflare is going and will develop code for; Pages will have limited improvements in the future).

Key Vaults

Findings ...

- Low Latency, Edge Compute.
- 1,200 rw/5-minutes (have to rethink large number of writes).

Data ...

- **key** (512B)
- **value** (5MB)
- **metadata** (1kB)

Keep in mind ...

- Metadata: can only see via API.
- This is where it's easy to hit the paywall limit.

D1 Databases

D1 is Cloudflare's managed, serverless database with SQLite's SQL semantics, built-in disaster recovery, and Worker and HTTP API access.

Findings ...

- 500 MB (free), 10GB (paid)
- 100 Columns/Table, Unlimited Rows

Pages

Findings ...

- Available on all plans.
- Unlimited bandwidth (Free)
- 500 deploys per month (Free)
- Cloudflare is recommending Pages in favor of Workers.

Findings (deprecation) ...

- Workers are low-latency and at the edge.
- Existing projects, use Pages. New projects, use workers.

Cloudflare Summary & Best Practices

Summary

- Cannot be **client-side** application.
- Manipulate DOM Elements.
- Add Cookies.
- Reroute/Redirect (**301 & 302 Status Codes**).
- **API**: keep in mind cache-buster, CORS, paging.
- Access to Domain, Path, and Parameters.
- GitHub Actions (build process) **VERSUS** Github via Settings (was able to not build).

Best Practices

- DEV Environment: Ensure it has paid access.
- Watch KV Read/Write Limits (use bulk where possible).
- Build processes add code (impacts run time).
- Cache-bust the 60-second cache.
- Use the module pattern (**import/export**)
- Use the extended worker lifetime (**ctx.waitUntil**)

Hackathon Summary & Best Practices

Summary

- Start from templates/boilerplates; save time.
- Keep Cloudflare organized; name Workers clearly.
- Use environment variables for API keys.
- Use the free tier limits wisely (KV limits, Worker request limits).
- Monitor limits using the Analytics tab.

